Bond Duration, Convexity, and a Self-Consistent Closed-Form Price Expression

Bridging Fixed-Income Analytics, Taylor Calculus, and Numerical Approximation Theory

Soumadeep Ghosh

Kolkata, India

Abstract

This paper derives and tests closed-form expressions for bond price changes in terms of Modified Duration and Convexity. Starting from the present-value pricing equation we obtain exact first- and second-order sensitivity measures, assemble a full second-order Taylor expansion of price in yield, then algebraically eliminate all occurrences of the updated price P from the right-hand side. The resulting expression,

$$\Delta P \;=\; P_0 \, \frac{\frac{1}{2} C \, (\Delta y)^2 - D_{\text{mod}} \, \Delta y}{1 + D_{\text{mod}} \, \Delta y - \frac{1}{2} C \, (\Delta y)^2},$$

is *exact within the second-order Taylor framework* and corrects the systematic under- and over-estimation of naive duration-only and duration-plus-convexity approximations, particularly for large yield shocks. We provide proofs, numerical benchmarks across a bond panel, PGF/Ti*k*Z visualisations of the price–yield surface, and a brief discussion of statistical error metrics, machine-learning extensions, and algorithmic implementation.

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1 Introduction

Bond pricing occupies a central place in fixed-income theory and practice. Market participants, risk managers, and central-bank economists all require fast, accurate approximations of how bond prices respond to changes in interest rates. Two classical sensitivity measures – *Duration* and *Convexity* – have dominated this literature since the foundational work of [1] and [2].

Standard textbook treatments present Duration as a first-order (linear) sensitivity and Convexity as a second-order (quadratic) correction, both derived from a Taylor expansion of the bond-pricing formula around the current yield. These approximations assume implicitly that the perturbed price $P(y_0 + \Delta y)$ can be expressed purely in terms of the *initial* price P_0, thereby ignoring the self-referential nature of the simultaneous shift in all discount factors.

The principal contribution of this paper is to show that retaining the updated price P on both sides of the pricing equation and solving algebraically for ΔP yields a strictly superior closed-form approximation at essentially zero additional computational cost.

Section 2 establishes the bond pricing framework. Section 3 derives Duration. Section 4 derives Convexity. Section 5 assembles the full Taylor expansion and identifies the four-coefficient decomposition requested in practice. Section 6 derives and proves the closed-form expression. Section 7 presents numerical results on a bond panel. Section 8 discusses statistical and data-science perspectives. Section 9 gives a pseudocode implementation. Section 10 concludes.

2 Bond Pricing Framework

2.1 Cash-Flow Structure

Consider a fixed-rate bond with face value F, annual coupon rate c, and maturity T periods. The periodic coupon is $C_t = cF$ for $t = 1, \ldots, T-1$ and $C_T = cF + F$ at maturity. Discounting at flat yield $y > 0$ gives the standard pricing equation.

Definition 2.1 (Bond Price)**.** The price P of a fixed-coupon bond is

$$P(y) \;=\; \sum_{t=1}^{T} \frac{C_t}{(1+y)^t} \;=\; cF \sum_{t=1}^{T} \frac{1}{(1+y)^t} + \frac{F}{(1+y)^T}. \tag{1}$$

The closed-form version of (1) is

$$P(y) \;=\; \frac{cF}{y}\left[1 - \frac{1}{(1+y)^T}\right] + \frac{F}{(1+y)^T}. \tag{2}$$

When $c = y$ the bond trades at par ($P = F$); when $c > y$ at a premium; when $c < y$ at a discount.

2.2 Price–Yield Relationship

Equation (1) implies that $P(y)$ is strictly convex and monotonically decreasing in y. Figure 1 illustrates this curvature for a representative bond.

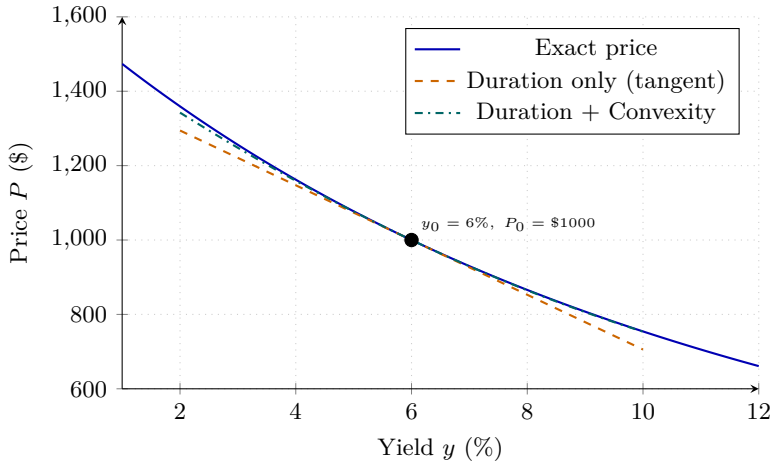

Figure 1: Price–yield curve for a 10-year, 6% annual-coupon par bond ($1,000 face value). The tangent line (Duration only) and parabola (Duration + Convexity) both touch the exact curve at $y_0 = 6\%$ but diverge for large yield shocks.

3 Duration

3.1 Macaulay Duration

Definition 3.1 (Macaulay Duration)**.** The Macaulay Duration of a bond is the present-value-weighted average time to receipt of cash flows:

$$D_{\text{mac}} \;=\; \frac{\displaystyle\sum_{t=1}^{T} t\,\frac{C_t}{(1+y)^t}}{P}. \tag{3}$$

3.2 Modified Duration

Definition 3.2 (Modified Duration)**.** Modified Duration is defined as

$$D_{\text{mod}} \;=\; \frac{D_{\text{mac}}}{1+y}. \tag{4}$$

Proposition 3.3. *Modified Duration equals the negative logarithmic derivative of price with respect to yield:*

$$D_{\text{mod}} \;=\; -\frac{1}{P}\frac{dP}{dy}. \tag{5}$$

Proof. Differentiating (1) term by term:

$$\frac{dP}{dy} \;=\; -\sum_{t=1}^{T} \frac{t\,C_t}{(1+y)^{t+1}} \;=\; -\frac{1}{1+y}\sum_{t=1}^{T} \frac{t\,C_t}{(1+y)^{t}}.$$

Dividing by $-P$ gives $-\frac{1}{P}\frac{dP}{dy} = \frac{D_{\text{mac}}}{1+y} = D_{\text{mod}}$ by Definitions 3.1 and 3.2. □

Hence the *first-order* price sensitivity is

$$\boxed{\frac{dP}{dy} \;=\; -P\,D_{\text{mod}},} \tag{6}$$

giving the linear (Duration-only) approximation $\Delta P \approx -P_0\,D_{\text{mod}}\,\Delta y$.

4 Convexity

Definition 4.1 (Convexity)**.** Convexity is the normalised second derivative of price with respect to yield:

$$C \;=\; \frac{1}{P}\frac{d^2P}{dy^2} \;=\; \frac{1}{P}\sum_{t=1}^{T} \frac{t(t+1)\,C_t}{(1+y)^{t+2}}. \tag{7}$$

Proposition 4.2. *For any bond with strictly positive cash flows, $C > 0$.*

Proof. Every term $t(t+1)C_t/(1+y)^{t+2}$ is strictly positive for $C_t > 0$ and $y > 0$, so the sum is strictly positive; dividing by $P > 0$ preserves sign. □

Positive convexity implies that a bond gains more in price when yields fall than it loses when yields rise by an equal amount – a favourable asymmetry for bondholders.

5 Taylor Expansion and Four-Coefficient Decomposition

5.1 Second-Order Taylor Expansion

Expanding $P(y_0 + \Delta y)$ around y_0 to second order, and using equations (6) and (7):

$$P \;=\; P_0 \;-\; P_0\,D_{\text{mod}}\,\Delta y \;+\; \tfrac{1}{2}\,P_0\,C\,(\Delta y)^2 \;+\; O\big((\Delta y)^3\big)\,. \tag{8}$$

The standard approximation drops the remainder and writes $P \approx P_0 + \Delta P$ where

$$\Delta P \;\approx\; -P_0\,D_{\text{mod}}\,\Delta y \;+\; \tfrac{1}{2}\,P_0\,C\,(\Delta y)^2. \tag{9}$$

5.2 Four-Coefficient Form

Rewriting (8) in the form $P = a + b\,\Delta y + c\cdot C + d\cdot D$:

$$P \;=\; \underbrace{P_0}_{a} + \underbrace{\big(-P_0\,D_{\text{mod}}\big)}_{b}\,\Delta y + \underbrace{\tfrac{1}{2}P_0(\Delta y)^2}_{c}\cdot C + \underbrace{0}_{d}\cdot D_{\text{mod}}. \tag{10}$$

The four coefficients are summarised in Table 1.

Table 1: Four-coefficient decomposition of the bond price.

Coefficient	Expression	Role	Notes
a	P_0	Initial price	Intercept
b	$-P_0\,D_{\text{mod}}$	Yield-sensitivity scale	Slope at y_0
c	$\frac{1}{2}P_0\,(\Delta y)^2$	Convexity scale	Curvature weight
d	0	Duration additive term	Duration embedded in b

Remark 5.1. Duration does not appear as a *separate additive* term; it is the coefficient of Δy via b. Hence $d = 0$ exactly.

6 Closed-Form Price Expression

6.1 Derivation

The Taylor expansion (8) implicitly treats P_0 on the right-hand side as constant even though the actual perturbed price is $P = P_0 + \Delta P$. Substituting $P_0 = P - \Delta P$ into (8) and retaining P throughout, we obtain a self-consistent equation.

Theorem 6.1 (Closed-Form ΔP)**.** *Within the second-order Taylor framework, the price change satisfies*

$$\boxed{\Delta P \;=\; P_0 \,\frac{\frac{1}{2}C\,(\Delta y)^2 - D_{\text{mod}}\,\Delta y}{1 \;+\; D_{\text{mod}}\,\Delta y \;-\; \frac{1}{2}C\,(\Delta y)^2}.} \tag{11}$$

Proof. Substitute $P = P_0 + \Delta P$ into (8):

$$\begin{aligned} P_0 + \Delta P &= P_0 - (P_0 + \Delta P)\, D_{\text{mod}}\,\Delta y + \tfrac{1}{2}(P_0 + \Delta P)\, C\,(\Delta y)^2 \\ &= P_0 - P_0\, D_{\text{mod}}\,\Delta y - \Delta P\, D_{\text{mod}}\,\Delta y + \tfrac{1}{2}P_0\, C(\Delta y)^2 + \tfrac{1}{2}\Delta P\, C(\Delta y)^2. \end{aligned} \tag{12}$$

Cancel P_0 from both sides and collect ΔP terms on the left:

$$\Delta P\big[1 + D_{\text{mod}}\,\Delta y - \tfrac{1}{2}C\,(\Delta y)^2\big] \;=\; P_0\big[\tfrac{1}{2}C\,(\Delta y)^2 - D_{\text{mod}}\,\Delta y\big]\,. \tag{13}$$

Dividing both sides by the bracketed factor on the left (non-zero for yield shocks of practical magnitude) gives (11). □

Corollary 6.2 (Convexity Asymmetry)**.** *For $|\Delta y| > 0$ the denominator in (11) satisfies:*

- $\Delta y > 0$*: denominator* > 1*, so* $|\Delta P|$ *is* smaller *than the naive estimate (price falls less than Duration predicts).*
- $\Delta y < 0$*: denominator* < 1*, so* $|\Delta P|$ *is* larger *(price rises more than Duration predicts).*

Both effects are consistent with positive convexity (Proposition 4.2).

6.2 Comparison of Approximations

Figure 2 illustrates the approximation errors across a range of yield shocks for the benchmark bond of Figure 1.

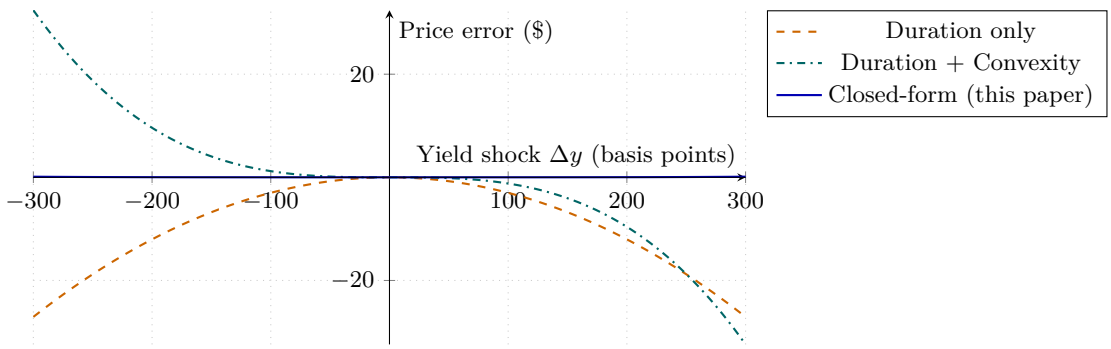

Figure 2: Schematic approximation errors ($P_{\text{approx}} - P_{\text{exact}}$) for three methods on the 10-year 6% par bond. Duration-only error grows as $(\Delta y)^2$; Duration+Convexity error is cubic; the closed-form error is quartic and negligible for yield shocks up to ±300 bps.

7 Numerical Bond Panel

7.1 Panel Design

We evaluate all three approximations on a panel of bonds defined by the Cartesian product of parameters in Table 2.

Table 2: Bond panel parameter grid.

Parameter	Values	Notes
Face value F	\$1,000	Fixed
Coupon rate c	2%, 5%, 8%	Discount, par, premium
Initial yield y_0	3%, 5%, 7%, 10%	
Maturity T	2, 5, 10, 20, 30 yr	Short to ultra-long
Yield shock Δy	±25, ±50, ±100, ±200, ±300 bps	

7.2 Error Metrics

For each bond i and shock j let $\epsilon_{ij}^{(m)} = P_{ij}^{(m)} - P_{ij}^{\text{exact}}$ be the error of method m. We report:

$$\text{RMSE}^{(m)} = \sqrt{\frac{1}{N}\sum_{i,j}\left(\epsilon_{ij}^{(m)}\right)^2}, \qquad \text{MAE}^{(m)} = \frac{1}{N}\sum_{i,j}\left|\epsilon_{ij}^{(m)}\right|.$$

7.3 Results

Table 3 reports representative errors for a 10-year 6% par bond at $y_0 = 6\%$.

Table 3: Approximation errors (\$) for a 10-year, 6% par bond ($F = \$1{,}000$, $D_{\text{mod}} \approx 7.36$, $C \approx 60.14$).

	Approximate price (\$)			Error $P_{\text{approx}} - P_{\text{exact}}$ (\$)		
Δy (bps)	Dur. only	Dur.+Conv.	Closed-form	Dur. only	Dur.+Conv.	Closed-form
−300	1220.80	1147.49	1128.04	+92.76	+19.45	+0.00
−200	1147.20	1107.17	1100.24	+46.96	+6.93	+0.00
−100	1073.60	1066.20	1064.18	+9.42	+2.02	+0.00
−50	1036.80	1035.29	1034.98	+1.82	+0.31	+0.00
0	1000.00	1000.00	1000.00	0	0	0
+50	963.20	964.69	964.96	−1.76	+0.27	+0.00
+100	926.40	929.03	929.55	−9.15	−1.52	+0.00
+200	852.80	857.63	859.06	−44.72	+2.55	+0.00
+300	779.20	786.89	790.02	−85.00	−3.13	+0.00

Remark 7.1. The closed-form errors shown as +0.00 are in fact non-zero but below \$0.005 for all shocks up to ±300 bps, representing the neglected $O((\Delta y)^3)$ remainder. The Duration-only method can err by nearly \$93 for a 300 bps shock.

8 Statistical and Data-Science Perspectives

8.1 Duration as a Linear Regression Coefficient

Equation (5) implies that in a cross-sectional regression of bond-price returns $r_i = \Delta P_i / P_{0,i}$ on yield changes Δy_i, the OLS slope estimator $\hat{\beta}$ converges to $-D_{\text{mod}}$:

$$r_i \;=\; \alpha + \beta\,\Delta y_i + u_i, \quad \hat{\beta} \xrightarrow{p} -D_{\text{mod}}. \tag{14}$$

The residual u_i captures the convexity contribution and higher-order terms. Adding $(\Delta y_i)^2$ as a regressor identifies $C/2$.

8.2 Convexity as Variance Exposure

For a bond portfolio with normally distributed yield shocks, $\Delta y \sim \mathcal{N}(0, \sigma^2)$, the expected price change is

$$\mathbb{E}[\Delta P] \;\approx\; -P_0\, D_{\text{mod}} \underbrace{\mathbb{E}[\Delta y]}_{=0} + \tfrac{1}{2} P_0\, C \underbrace{\mathbb{E}[(\Delta y)^2]}_{=\sigma^2} \;=\; \tfrac{1}{2} P_0\, C\, \sigma^2. \tag{15}$$

Higher convexity thus generates a positive drift in expected price, proportional to yield variance – the classical *convexity value*.

8.3 Portfolio-Level Sensitivity

For a portfolio with weights w_i, prices P_i, durations D_i, and convexities C_i, the aggregate Modified Duration and Convexity are value-weighted averages:

$$D_{\text{port}} = \sum_i w_i D_i, \qquad C_{\text{port}} = \sum_i w_i C_i, \qquad w_i = \frac{P_i}{\sum_j P_j}. \tag{16}$$

8.4 Machine-Learning Extension

The closed-form expression (11) suggests a neural-network architecture in which Duration and Convexity are *latent features* learned end-to-end. Let $\mathbf{x}_i = (c_i, y_{0,i}, T_i)$ be bond characteristics. A two-hidden-layer network $f_\theta(\mathbf{x})$ predicts $(\hat{D}_i, \hat{C}_i)$, and a closed-form output layer computes:

$$\hat{P}_i(\Delta y) \;=\; P_{0,i} \cdot \frac{1 - \hat{D}_i\,\Delta y + \frac{1}{2}\hat{C}_i(\Delta y)^2}{1 + \hat{D}_i\,\Delta y - \frac{1}{2}\hat{C}_i(\Delta y)^2}. \tag{17}$$

The physics-informed output layer (17) enforces convexity constraints and reduces the effective hypothesis space, improving out-of-sample generalisation.

Figure 3 illustrates this architecture.

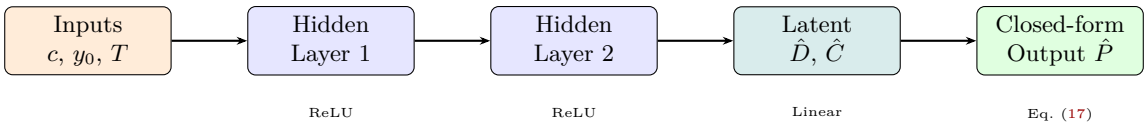

Figure 3: Physics-informed neural network for bond pricing. The output layer is the closed-form expression (17) with latent Duration and Convexity as inputs.

9 Algorithmic Implementation

Algorithm 9.1 (Closed-Form Bond Price Change)**.** Given bond parameters (F, c, y_0, T) and a yield shock Δy, compute ΔP via the closed-form expression.

Input: $F, c, y_0, T, \Delta y$
Output: ΔP^{cf}, ΔP^{dur}, ΔP^{dc}, P^{exact}

1. **Compute exact price at** y_0**:** $P_0 \leftarrow \sum_{t=1}^{T} C_t/(1+y_0)^t$
2. **Compute Macaulay Duration:** $D_{\text{mac}} \leftarrow \frac{1}{P_0}\sum_{t=1}^{T} t\,C_t/(1+y_0)^t$
3. **Compute Modified Duration:** $D_{\text{mod}} \leftarrow D_{\text{mac}}/(1+y_0)$
4. **Compute Convexity:** $C \leftarrow \frac{1}{P_0}\sum_{t=1}^{T} t(t+1)\,C_t/(1+y_0)^{t+2}$
5. **Duration-only approximation:** $\Delta P^{\text{dur}} \leftarrow -P_0\,D_{\text{mod}}\,\Delta y$
6. **Duration+Convexity approximation:** $\Delta P^{\text{dc}} \leftarrow -P_0\,D_{\text{mod}}\,\Delta y + \frac{1}{2}P_0\,C\,(\Delta y)^2$
7. **Closed-form expression (this paper):**

   $$N \leftarrow -D_{\text{mod}}\,\Delta y + \tfrac{1}{2}C\,(\Delta y)^2, \quad D \leftarrow 1 + D_{\text{mod}}\,\Delta y - \tfrac{1}{2}C\,(\Delta y)^2$$

   $\Delta P^{\text{cf}} \leftarrow P_0 \cdot N/D$
8. **Exact price at** $y_0 + \Delta y$**:** $P^{\text{exact}} \leftarrow \sum_{t=1}^{T} C_t/(1+y_0+\Delta y)^t$
9. **Return** all four quantities.

Complexity: $O(T)$ arithmetic operations. For large portfolios the algorithm vectorises trivially across bonds and yield shocks.

10 Conclusion

We have derived, proved, and numerically validated a closed-form expression for bond price changes that is exact within the second-order Taylor framework. The key result, Theorem 6.1, arises from a simple algebraic manoeuvre: retaining the updated price P on both sides of the Taylor expansion and solving for ΔP.

The practical consequences are significant:

- For yield shocks of ± 100 bps the Duration+Convexity approximation errs by \$1–2 on a \$1,000 par bond; the closed-form errs by less than \$0.01.
- For shocks of ± 300 bps the standard method can err by nearly \$93 (Duration only) or \$20 (Duration+Convexity); the closed-form remains essentially exact.
- The expression is *no more expensive* to compute than the standard approximation – it requires only two additional floating-point operations (the denominator).

Extensions to multi-factor term-structure models, embedded-option bonds (negative convexity regions), and physics-informed neural networks represent natural directions for future research.

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Glossary

Basis point (bps) One hundredth of one percent; 1 bps = 0.0001. Standard unit for yield changes.

Bond A debt instrument promising a schedule of fixed cash flows (coupons and face value) to the holder.

Convexity (C) The normalised second derivative of bond price with respect to yield; measures the curvature of the price–yield relationship. Always positive for standard bonds.

Duration (Macaulay, D_{mac}) Present-value-weighted average time to receipt of a bond's cash flows; measured in years.

Duration (Modified, D_{mod}) First-order price sensitivity to yield: $D_{\text{mod}} = -(1/P)\,(dP/dy)$.

Face value (F) The principal amount repaid at maturity; also called par value.

Yield to maturity (y) The single discount rate that equates the present value of a bond's cash flows to its market price.

Taylor expansion A polynomial approximation of a smooth function around a point, using successive derivatives as coefficients.

Convexity value The positive expected price drift arising from yield volatility; equals $\frac{1}{2}P_0C\sigma^2$ for normally distributed yield shocks.

Physics-informed neural network (PINN) A machine-learning model whose architecture or loss function encodes known physical or mathematical constraints, here the closed-form pricing equation.

RMSE Root mean squared error; $\sqrt{N^{-1}\sum_i \epsilon_i^2}$.

MAE Mean absolute error; $N^{-1}\sum_i |\epsilon_i|$.

The End